THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 15 units	$\sqrt{81+144}=15$
2	**(d)** $(0,\,-5)$	$x=0$, and below the origin means $y<0$
3	**(c)** $(2,\,2)$	$\left(\frac{-3+(7)}{2},\frac{5+(-1)}{2}\right)$
4	**(d)** 4 and -7	compare with $y=ax+b$
5	**(a)** False, its degree is 2 so it is quadratic	degree of $5x^2-3$ is 2
6	**(b)** 7	$\lvert-4-3\rvert=7$
7	**(a)** irrational	rational + irrational
8	**(b)** $\frac{3}{5}$	distance from 0
9	**(d)** 11025	$105^2=(100+5)^2$
10	**(c)** $(x+y)^2=x^2+2xy+y^2$	true for all values of the variables
11	**(b)** 30 cm	half-chord $=\sqrt{17^2-8^2}=15$, so chord $=2\times15=30$
12	**(a)** the centre of the circle	centre is equidistant from the chord's endpoints
13	**(d)** 17 cm	$r=\sqrt{15^2+8^2}=17$
14	**(d)** 132 cm	$2\pi r=2\times\frac{22}{7}\times21=132$
15	**(a)** 60 cm^2	$s=20$, area $=\sqrt{20\cdot12\cdot5\cdot3}=60$
16	**(d)** approximate (approaching)	he realised the ratio could not be exact
17	**(c)** 12	multiply the number of outcomes of each experiment
18	**(d)** 0.15	$6/40=0.15$
19	**(b)** 98	$a+(n-1)d=2+24\times(4)=98$
20	**(c)** $n=37$	$7n+2=261\Rightarrow n=37$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$AB=\sqrt{4+4}=2\sqrt{2},\ BC=2\sqrt{2},\ AC=\sqrt{16+16}=4\sqrt{2}$. Since $AB+BC=AC$, the points are collinear.	1+1
22	$y=2x+3$; it cuts the y-axis at $(0,3)$.	1+1
23	Average $=\frac{5}{8}$; average of $\frac{1}{2}$ and $\frac{5}{8}=\frac{9}{16}$. So $\frac{9}{16}$ and $\frac{5}{8}$ (any two valid).	1+1
24	LHS $=27$; RHS $=8+12+6+1=27$ ✓.	1+1
25	$\angle BCD=180^\circ-70^\circ=110^\circ$; opposite angle $\angle A=180^\circ-110^\circ=70^\circ$.	1+1
26	Height $=\sqrt{25-9}=4$ cm; area $=\frac{1}{2}\times6\times4=12$ cm^2.	1+1
27	7 months (Jan, Mar, May, Jul, Aug, Oct, Dec) out of 12: $\frac{7}{12}$.	1+1
28	₹240, ₹280, ₹320 — an AP with $d=40$ (1). $t_n=200+40n$; $t_{10}=$ ₹600 (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$P=\left(\frac{2(2)+8}{3},\frac{2(-2)+4}{3}\right)=(4,0)$. $AP=\sqrt{4+4}=2\sqrt{2}$; $AB=\sqrt{36+36}=6\sqrt{2}$; $AP=\frac{1}{3}AB$ ✓	2+1
30	Points: $(0,0),(4,2)$; $(0,0),(2,2)$; $(0,0),(2,4)$. All pass through the origin; larger a ⇒ steeper line; $a=1$ makes equal angles with both axes.	2+1
31	$\sqrt{1^2+1^2}=\sqrt{2}$; then legs $\sqrt{2},1\Rightarrow\sqrt{3}$; then legs $\sqrt{3},1\Rightarrow\sqrt{4}=2$.	1+1+1
32	Area $=x^2+8x+15=(x+3)(x+5)$: sides $x+5$ and $x+3$.	2+1
33	A triangle is a cyclic 4-gon with $d=0$ (1). Then $s=\frac{a+b+c}{2}$ and $\sqrt{(s-a)(s-b)(s-c)(s-0)}=\sqrt{s(s-a)(s-b)(s-c)}$ (2), Heron's formula.	1+2
34	Subjective: a person's own opinion from what they see ('sunny, so unlikely'), differing between people (1). Objective: based on evidence such as past rainfall data (relative frequency) or on equally likely outcomes (theoretical) (2).	1+2

35	$\angle BDC = \angle BAC = 30^\circ$ (angles in the same segment) (1+1). $\angle CBD = 180^\circ - 100^\circ - 30^\circ = 50^\circ$ (1).	2+1
	OR: Central angle is double: $6x - 10 = 2(2x + 5) = 4x + 10$ (2) ⇒ $x = 10$; angles 50° and 25° (1).	2+1
36	$t_n + t_{n-1} = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} = n^2$ (2). $55 + 45 = 100 = 10^2$ ✓ (1).	2+1
	OR: $t_n = 3n - 1$ is linear: points $(1, 2), (2, 5), (3, 8)$ lie on $y = 3x - 1$ (2). Slope $= 3$, the common difference (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{1}{4} - \frac{5}{12} = \frac{3-5}{12} = -\frac{1}{6}$; $-\frac{1}{6} \div \frac{2}{3} = -\frac{1}{6} \times \frac{3}{2} = -\frac{1}{4}$. (ii) Sum $= \frac{6}{10} + \frac{7}{10} = \frac{13}{10}$; product $= \frac{21}{50}$.	3+2
	OR: (i) $\frac{x}{2} = \frac{9}{2} \Rightarrow x = 9$. (ii) LHS $= \frac{1}{2} \times \frac{13}{12} = \frac{13}{24}$; RHS $= \frac{3}{8} + \frac{1}{6} = \frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ ✓	2+3
38	Opposite angles of a parallelogram are equal and in a cyclic quadrilateral they add to 180°, so each is 90°: a rectangle (3). Each diagonal then subtends 90° at the other two vertices, so it is a diameter (2).	3+2
	OR: Let $OM = ON$ with $OM \perp AB, ON \perp CD$. Right triangles OMA, ONC: $OA = OC$, $OM = ON$ ⇒ congruent (RHS) ⇒ $AM = CN$ (2). Halves equal ⇒ $AB = CD$ (1). Length $= 2\sqrt{169 - 144} = 10$ cm (2).	2+1+2
39	(i) $\frac{22}{7} \times 1225 = 3850$ m^2 (1). (ii) $\frac{22}{7} \times 49 = 154$ m^2 (1). (iii) $3850 - 154 = 3696$ m^2 (1). (iv) $2\pi(35) + 2\pi(7) = 220 + 44 = 264$ m (2).	1+1+1+2
	OR: (i) $\frac{360}{8} = 45^\circ$ (1). (ii) $\frac{45}{360} \times 2\pi \times 14 = 11$ cm (1). (iii) $11 + 14 + 14 = 39$ cm (1). (iv) $\frac{45}{360} \times \frac{22}{7} \times 196 = 77$ cm^2 (2).	1+1+1+2
40	(a) AP, $d = 4$, $t_n = 4n + 3$ (1). (b) GP, $r = 3$, $t_n = 2 \cdot 3^{n-1}$ (1). (c) Neither (differences 3,5,7; ratios 4, 2.25) but $t_n = n^2$ (1). (d) GP, $r = \frac{1}{2}$, $t_n = 100\left(\frac{1}{2}\right)^{n-1}$ (2).	1+1+1+2
	OR: (i) Points $(1, 1), (2, 5), (3, 9), (4, 13), (5, 17)$ lie on the line $y = 4x - 3$ (2). (ii) $(1, 3), (2, 6), (3, 12), (4, 24), (5, 48)$ do not lie on a line; slopes 3, 6, 12, 24 keep rising (2). (iii) An AP grows by a fixed amount (linear), a GP by a fixed multiple (much faster) (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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